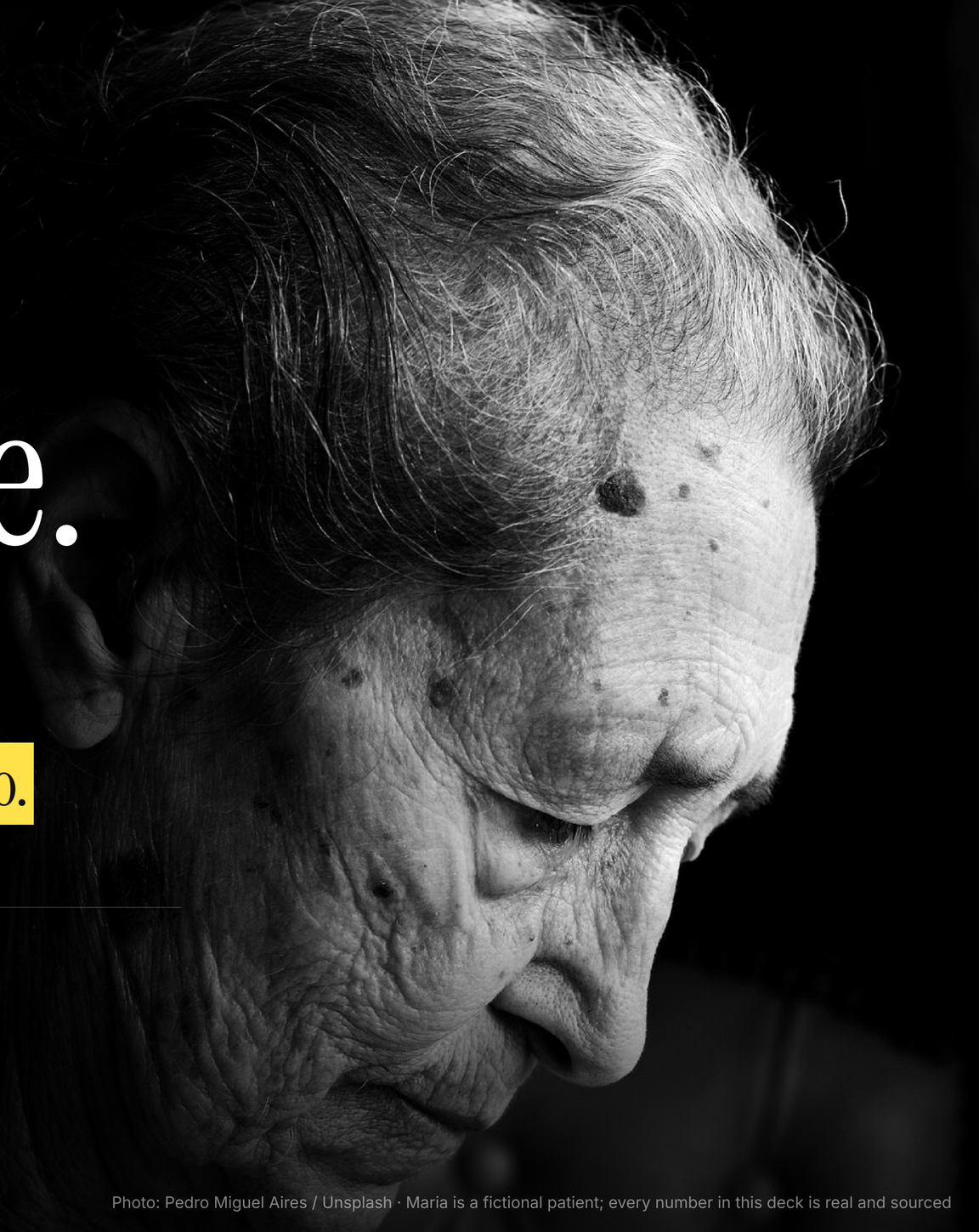
ITALY · OCTOBER 2026

On Friday, Maria goes home.

She is 84. She fell in her hallway at night and broke her hip. She will go home with a walker.

No one has ever seen the home she is going back to.



Step Wise

Home fall-hazard assessment from phone photos, reviewed by a professional before discharge.

AMD Developer Hackathon · ACT III
Track 2 · Health & Wellbeing

Photo: Pedro Miguel Aires / Unsplash · Maria is a fictional patient; every number in this deck is real and sourced

THE PROBLEM

At work, safety is mandatory. At home, it is not.

At work · Legislative Decree 81/2008

- Mandatory risk assessment
- A designated safety officer
- Inspections and penalties

1,189 deaths in 2025

inail.it/content/dam/inail-hub-site/documenti/2026/09/relazione-annuale-inail-2025-nota-stampa-infortuni-mp.pdf

INAIL NOTA PER LA STAMPA

diminuzione del 3,3% rispetto ai 1.239 del 2024 (41 casi in meno). Nel dettaglio si registrano 1.189 denunce mortali per i lavoratori (37 in meno rispetto all'anno precedente) e nove per

INAIL · Annual report 2025 ORIGINAL SOURCE

At home · no rules

- ~~Risk assessment~~
- ~~Safety officer~~
- ~~Inspections~~

~6,000 deaths every year · each square = 10

epicentro.iss.it/cong_leb/balducci

epicentro Il portale dell'epidemiologia per la sanità pubblica

Indice A-Z Cerca

In Italia ogni anno muoiono circa 6000 persone per incidente in casa, considerando come tale sia

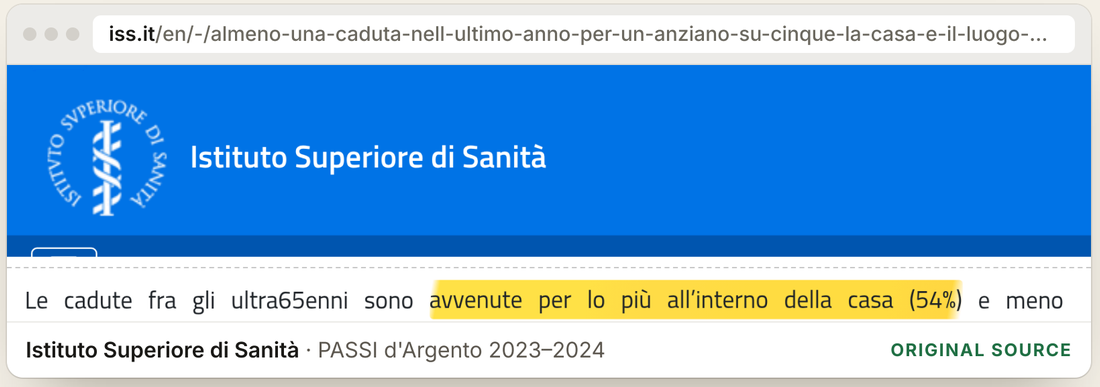
Istituto Superiore di Sanità · Deaths from home accidents ORIGINAL SOURCE

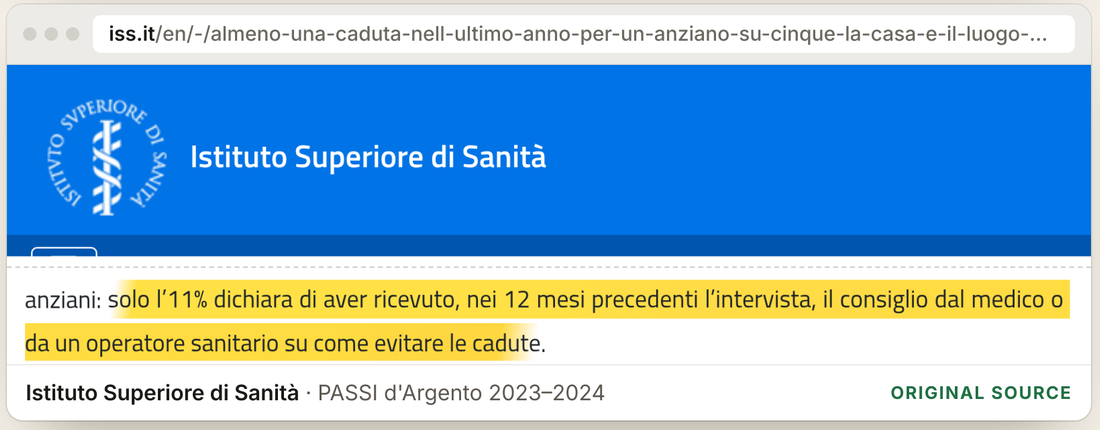
WHO FALLS, AND WHERE

54%

of falls among people over 65 happen at home.

11% were advised by a doctor on how to avoid them.



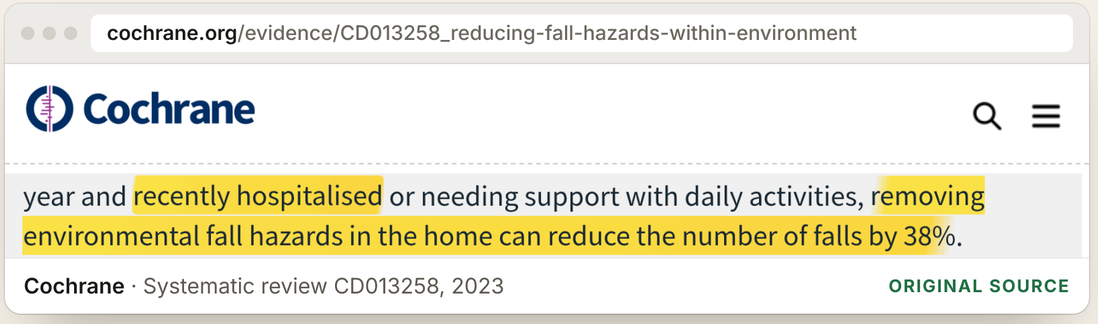


PREVENTION WORKS

−38%

falls when an expert inspects the home and hazards are removed, in high-risk older adults: like someone just discharged from hospital.

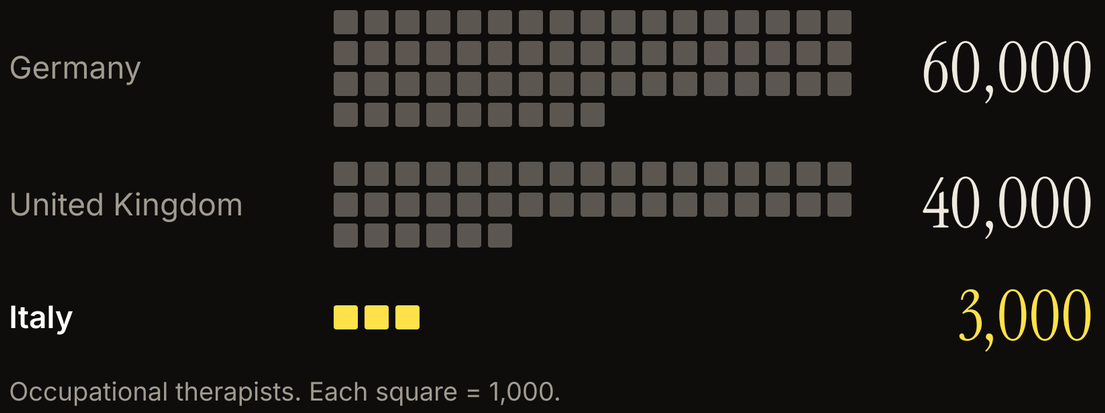
That expert is the occupational therapist.



Cochrane systematic review 2023: 22 studies, 8,463 people, high-certainty evidence.

THE BOTTLENECK

In Italy, that expert is missing.



Occupational therapists. Each square = 1,000.

1 therapist for every ~5,000 people over 65.

tsrm-pstrp.org/index.php/la-giornata-nazionale-del-terapista-occupazionale-2024/

FNO TSRM e PSTRP
Federazione nazionale degli Ordini dei tecnici sanitari di radiologia medica e delle professioni sanitarie tecniche, della riabilitazione e della prevenzione

Sono circa 3 mila i professionisti sanitari iscritti agli albi dei Terapisti occupazionali -

FNO TSRM e PSTRP · Occupational therapists in Italy, 2024 ORIGINAL SOURCE

tsrm-pstrp.org/index.php/la-giornata-nazionale-del-terapista-occupazionale-2024/

FNO TSRM e PSTRP
Federazione nazionale degli Ordini dei tecnici sanitari di radiologia medica e delle professioni sanitarie tecniche, della riabilitazione e della prevenzione

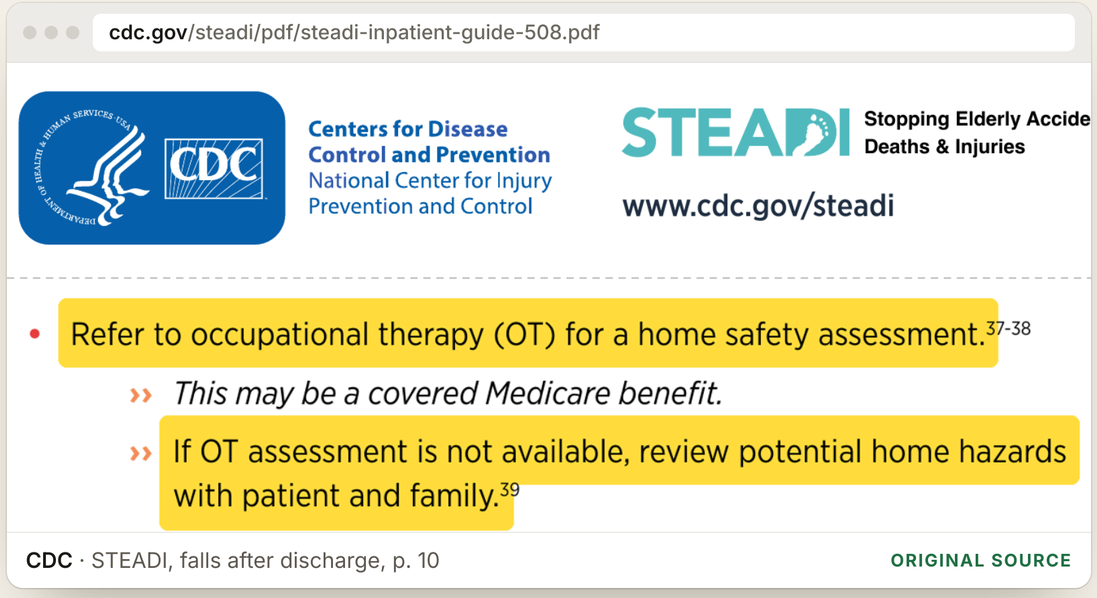
strumenti e modelli specifici di Terapia occupazionale. Ne danno conferma i numeri: in Germania sono oltre 60 mila, nel Regno Unito 40 mila, in Danimarca 11 mila, in Belgio

FNO TSRM e PSTRP · Occupational therapists in Europe, 2024 ORIGINAL SOURCE

THE MISSING RULE

«If an occupational therapy assessment is not available, review potential home hazards with the patient and family.»

The CDC recommends it at every discharge. **Step Wise makes it possible, with an expert eye.**



THE SOLUTION

From phone photos to a plan the family can act on.

Three AI models do the slow inspection. The professional has the last word.

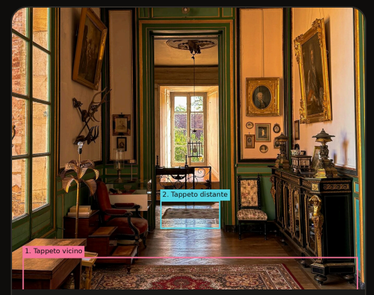


01

Capture

A relative photographs the home with a phone. The nurse adds the patient's profile.

`FAMILY · NURSE`



02

Detection

Jev answers atomic questions: is there a rug, a cable, a bathtub? Each with its box.

`TYPESAFE JEV`



03

3D room

VGGT turns the photos into a room the reviewer can walk through.

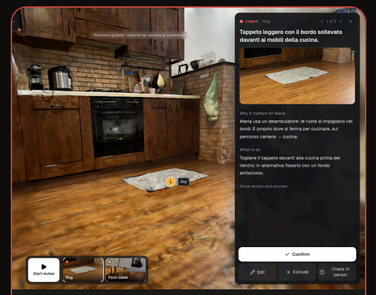
`VGGT-1B`

04

Reasoning

Gemma ranks the hazards and explains why each matters for this patient.

`GEMMA 4 31B`

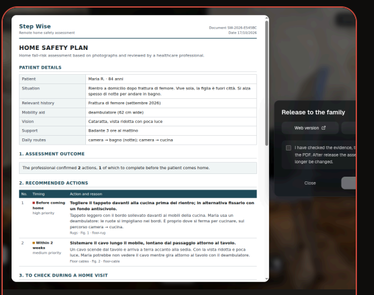


05

Review

The discharge nurse confirms, edits, excludes or defers each finding.

`PROFESSIONAL`



06

Family plan

Approved actions, by priority, in plain language, on the family's phone.

`FAMILY`

01 CAPTURE · 03 3D RECONSTRUCTION

From flat photos to a room.

VGGT is a transformer that looks at all the photos at once. In a single pass it works out where each one was taken from, how far away every pixel is, and where it sits in space.

1. **Camera poses**: where each photo was taken
2. **Depth**: the distance of every pixel
3. **3D points**: the room, with each hazard in its place

`VGGT-1B · Meta, CVPR 2025` `Gaussian splatting · gsplat` `PyTorch · ROCm`



↓ 39 phone photos



Real output: 39 phone photos of a real home, camera poses from VGGT-1B, 757,792 Gaussians trained with gsplat. The photographs remain the evidence; the 3D view helps the reviewer navigate. Scale not yet calibrated.

02 DETECTION · 04 REASONING

Every hazard, with its evidence.

Jev answers atomic questions on the areas of the **Westmead Home Safety Assessment**. A box counts only when presence is confirmed; what a photo cannot show stays “to check in person”.





Live Jev 1.14 calls. Photos: Thomas Würth, Point3D Commercial Imaging Ltd. / Unsplash.

Why it matters, for Maria.

Gemma reads the evidence and the confirmed profile, ranks the hazards by urgency and explains why. Every claim must point to a photo and a guideline.

CONFIRMED PROFILE

Walker · hip fracture · lives alone · gets up at night

EVIDENCE

The hazards found in the photos, each with its box

GUIDELINES

CDC STEADI

Gemma 4 31B

Google DeepMind on AMD

PRIORITY

Urgent, before coming home · or within weeks

WHY, FOR THIS PATIENT

“The walker's wheels catch on the rug's edge, on her daily route.”

WHAT TO DO

A plain-language action, linked to its photo and source

05 REVIEW

The professional decides.

The discharge nurse sees each hazard with its photo, its place in the room and the reason. Minutes, not a home visit.

Confirm
goes into the family plan

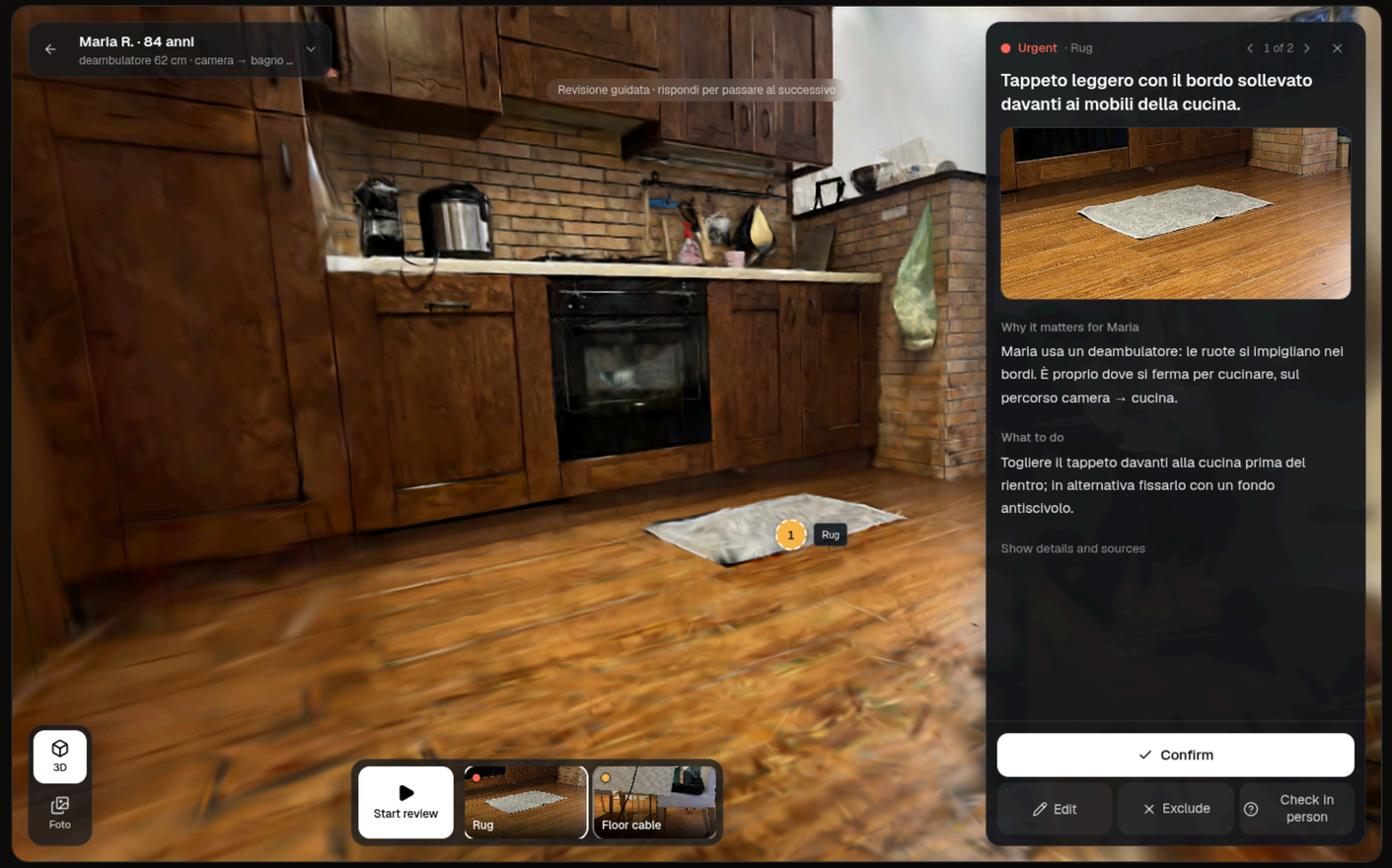
Edit
change the action or the wording

Exclude
never reaches the family

Check in person
left for the first home visit

Nothing reaches the family without her sign-off.

`Next.js · three.js` `versioned reviews` `release blocked if a decision changes`



Real prototype screen: guided review inside the room reconstructed from 39 photos. In this demo case the two hazards were marked by hand by the team.

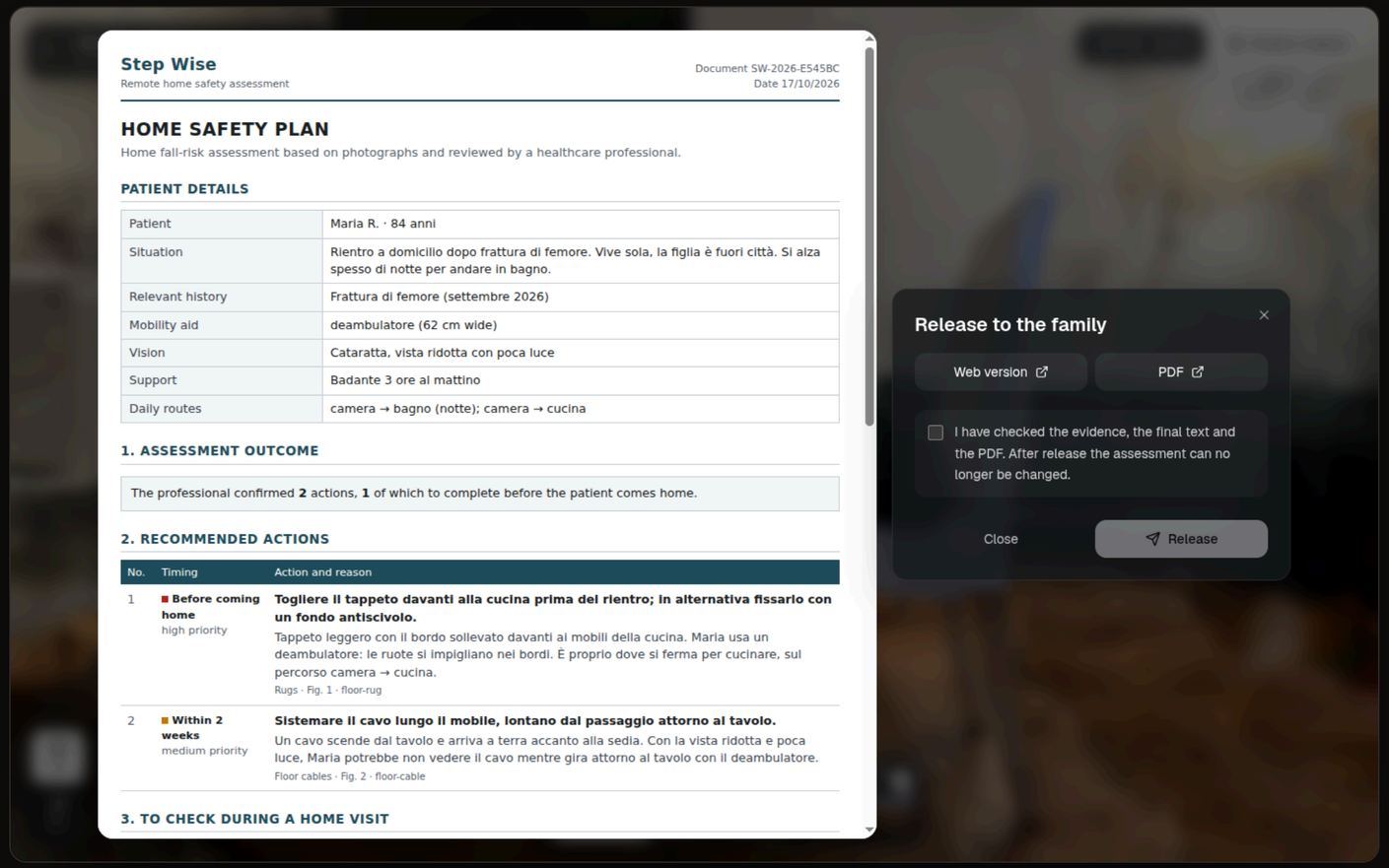
06 FAMILY PLAN

The family knows what to do.

Maria's daughter gets the approved plan on her phone: actions in order of priority, in plain language, each with its photo. Before Maria comes home, not after the next fall.

1. **Only confirmed hazards** enter the plan
2. **What the photos cannot assess** is listed for the home visit
3. **Private link**: expires in 7 days, can be revoked
4. **Same document** as web page and PDF, with a fingerprint

`Jinja2 · WeasyPrint` `HTML + PDF`



Real prototype screen: the generated home safety plan, released to the family with a private link.

WHY AMD

One GPU, inside the hospital.

Home photos and clinical data cannot go to just any cloud. The 192 GB of one MI300X hold both the 3D reconstruction and the model that reasons about the patient: everything stays inside the hospital.

01 · CAPTURE

Photos + profile

Upright orientation, metadata stripping, quality checks.

`FastAPI · Pillow · NumPy`

03 · RECONSTRUCTION

VGGT-1B

Camera poses, depth and 3D points.

`PyTorch · ROCm`

04 · REASONING

Gemma 4 31B

Checks, priorities, plan.

`vLLM · ROCm`

AMD Instinct MI300X · 192 GB HBM3

`ROCm · reconstruction and model on the same GPU · one job at a time`

02 · Detection: Jev 1.14, external API, receives only prepared photos with no patient data · 05–06 · Review and report: Next.js, three.js, WeasyPrint

192 GB	2 models	0	ROCm
of HBM3 memory on a single GPU	VGGT and Gemma on the same card	clinical data leaving the hospital	open stack: PyTorch and vLLM

WHO IT IS FOR

Built around the discharge, not around the AI.

USES IT

The discharge nurse

Reviews a home in minutes, from the ward, before the patient leaves. Decides what goes to the family.

BENEFITS

The patient and the family

A clear plan before coming home. A relative takes the photos; the same relative receives the actions.

ADOPTS IT

Hospitals and local health services

Their discharge pathway gains the home review the CDC recommends, on hardware they control.

Fills a real gap

With one occupational therapist for every ~5,000 people over 65, the real alternative to a home visit is no assessment at all.

Uses the scarce expert well

“Check in person” sends to a home visit only what the photos cannot settle.

Fits clinical rules

A professional signs off every finding; data stays inside the hospital.

THE PROTOTYPE TODAY

What works, and what comes next.

Working

End-to-end flow: case → 3D room → guided review → report → release → revocable family link

Real 3D room from 39 phone photos: VGGT-1B poses, 757,792 Gaussians

Live object checks with TypeSafe Jev 1.14, with presence and box gates

Gemma 4 31B wired to AMD's endpoint for hazard sweep and report drafting

Public judge demo: one frozen, pre-analysed case, a private copy per visitor

Next

Self-hosted pipeline on one AMD Instinct MI300X with ROCm

Calibrated scale and validated 3D hazard positions on real homes

Guided capture to help a relative take the right photos

Clinical pilot with a discharge unit: accuracy and reviewer time

Step Wise is a hackathon prototype, **not a medical device**

CODE

`github.com/jacopopper/amd-hack`

TEAM

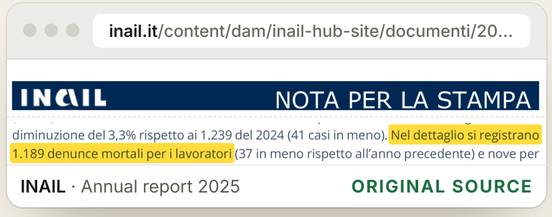
`@jacopopper · @davideblesse`

On Friday, Maria goes home.

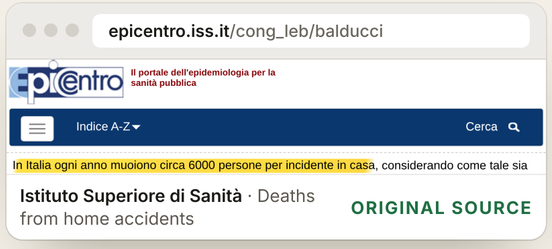
This time, someone looked at the hallway.

SOURCES

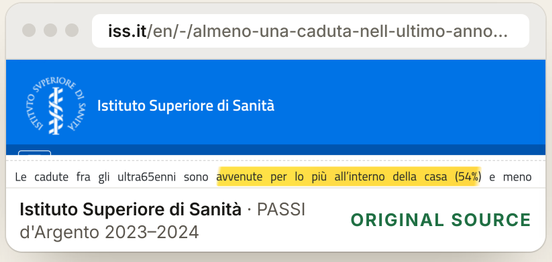
Every number comes from a public institution.



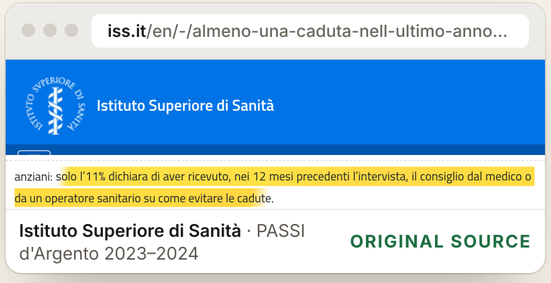
INAIL: public body overseen by the Italian Ministry of Labour.
https://www.inail.it/content/dam/inail-hub-site/documenti/2026/09/relazione-annuale-inail-2025-nota-stampa-infortuni-mp.pdf



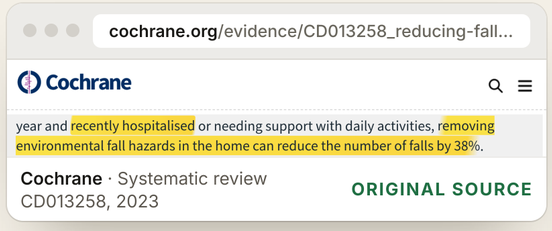
Istituto Superiore di Sanità: technical and scientific body of the Italian National Health Service.
https://www.epicentro.iss.it/cong_leb/balducci



Istituto Superiore di Sanità: national surveillance of people over 65.
https://www.iss.it/en/-/almeno-una-caduta-nell-ultimo-anno-per-un-anziano-su-cinque-la-casa-e-il-luogo-dove-si-verificano-di-piu



Istituto Superiore di Sanità: national surveillance of people over 65.
https://www.iss.it/en/-/almeno-una-caduta-nell-ultimo-anno-per-un-anziano-su-cinque-la-casa-e-il-luogo-dove-si-verificano-di-piu



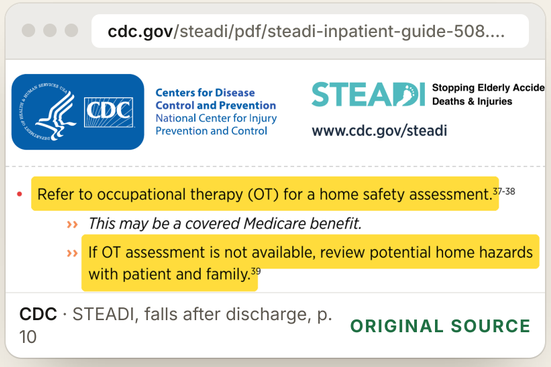
Cochrane: international reference for evidence-based medicine.
https://www.cochrane.org/evidence/CD013258_reducing-fall-hazards-within-environment



FNO TSRM e PSTRP: Federation of professional orders, overseen by the Italian Ministry of Health.
https://www.tsrm-pstrp.org/index.php/la-giornata-nazionale-del-terapista-occupazionale-2024/



FNO TSRM e PSTRP: Federation of professional orders, overseen by the Italian Ministry of Health.
https://www.tsrm-pstrp.org/index.php/la-giornata-nazionale-del-terapista-occupazionale-2024/



CDC: US federal public health agency.
https://www.cdc.gov/steadi/pdf/steadi-inpatient-guide-508.pdf